Evaluating Barometric Tactile Sensor Designs in Simulation

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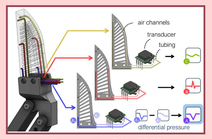
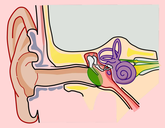
Challenge: Exploring barometric sensor designs is hard.

Barometric tactile sensors are gaining popularity, but **air channel designs remain simple**, especially compared to biological analogues



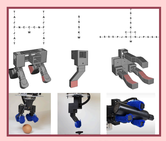
Shang et al., RAL, 2026

V.S.



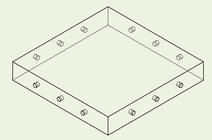
Human cochlea (Wikimedia Commons)

Computational design tools can explore a wider design space by **enabling faster design iteration**.

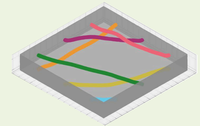


Zlokapa et al., ICRA 2022

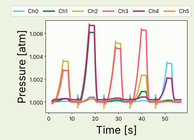
Our contribution: A design evaluation tool.



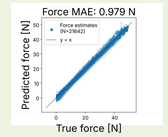
Target Object



Sensor Design



Simulated Behavior

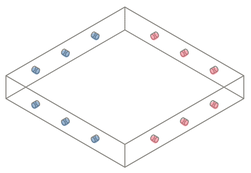


Estimator Performance

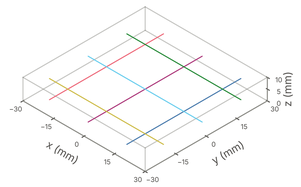
Our approach: For each sensor, we...

1. Generate a design

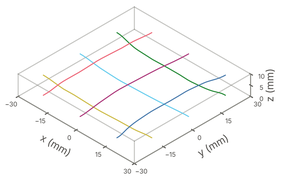
a Annotate base component in CAD



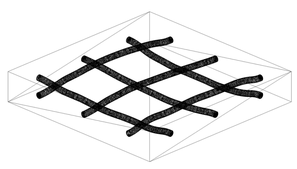
b Route each channel individually



c Optimizer resolves intersections

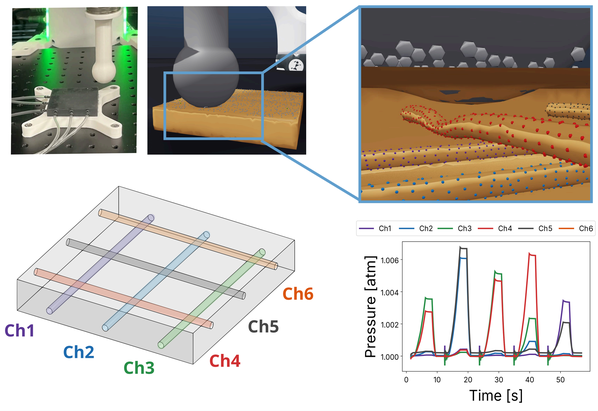


d Export fabrication-ready parts



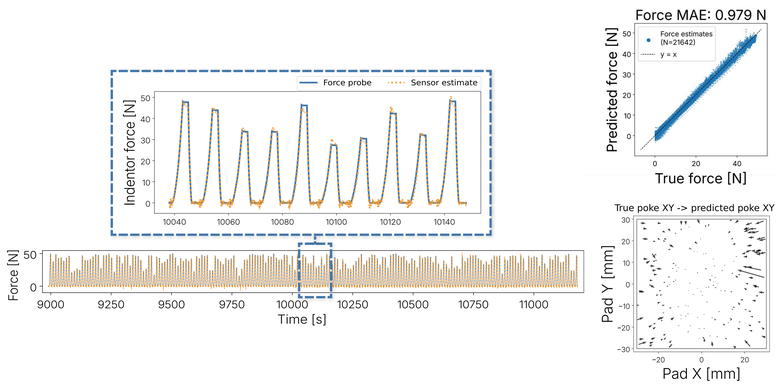
2. Simulate its response to pokes

To characterize a sensor pad design's performance, we simulate each design as a soft body in Nvidia Newton, and record the air pressure over one thousand pokes.

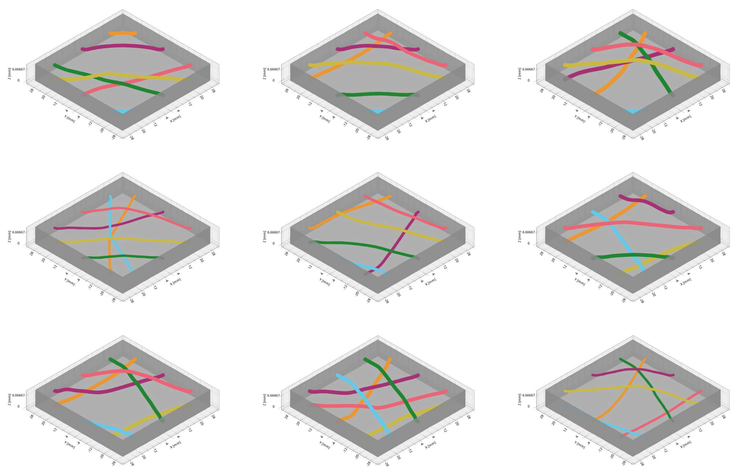


3. Train a force estimator

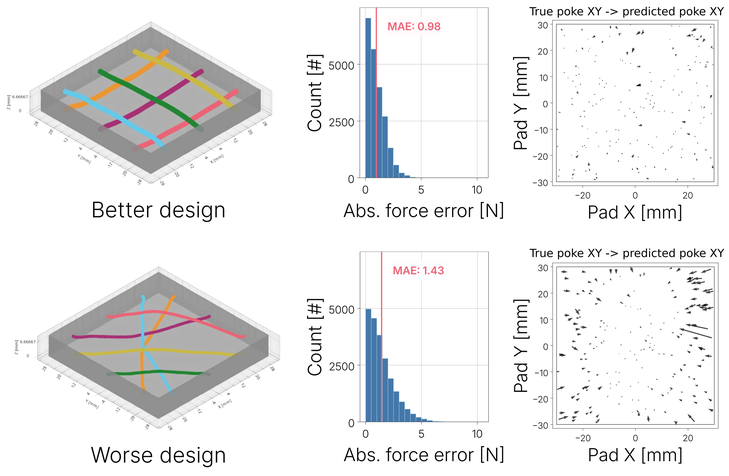
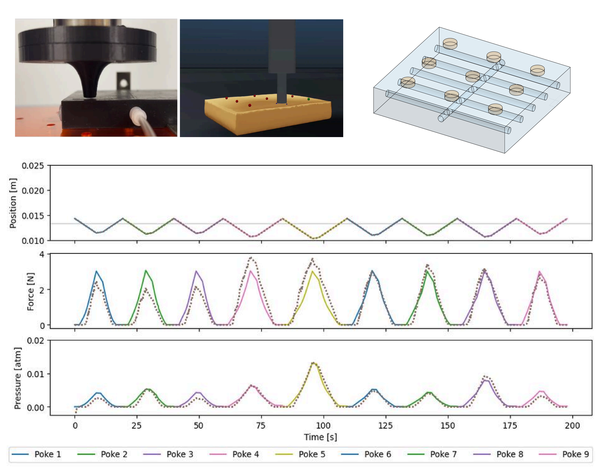
We train an RNN estimator to estimate the force norm, contact location, and uncertainty from the simulated 1k poke dataset. This estimator's performance determines the sensor design's overall performance.



Generated designs are diverse



Comparing sensors w/o fabrication



Sensor simulation matches reality